

Tracker

Webcam-Based Multi-Track Amplitude Modulation Player

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1 Overview

Tracker is a webcam-based multi-track amplitude modulation player. It tracks colored balls in front of a webcam and maps their positions to the volume of looping audio tracks using SDL2_mixer, with perspective-correction for angled camera setups.

Each ball controls two audio loops via its perspective-corrected position:

- **Ball 1 (red):** X-axis $\rightarrow$ Track 1 volume, Y-axis $\rightarrow$ Track 2 volume
- **Ball 2 (green):** X-axis $\rightarrow$ Track 3 volume, Y-axis $\rightarrow$ Track 4 volume
- Additional balls follow: ball $n \rightarrow$ tracks $2n$ and $2n + 1$
- All tracks loop continuously; volume is modulated in real time by ball position
- If a ball is lost, its tracks go **silent** (volume $\rightarrow 0$)

2 Perspective Correction

The webcam view may be angled, foreshortening ball positions. Define a planar region with 4 draggable corner handles; ball positions are rectified through a homography before mapping to volume.

- **D**rag the 4 corner handles (red = X axis, blue = Y axis) to outline the tracking plane
- **S** – save calibration to `calib.txt` (auto-loaded next run)
- **R** – reset corners to the full frame
- A perspective grid + quadrant crosshair is overlaid to show the rectification

3 Usage / Arguments

```
./tracker [N]
```

- N – number of balls to track (default 2, max 16)
- Example: `./tracker 3` tracks three balls, controlling tracks 1–6

Trackers auto-initialize at startup at staggered positions; lost trackers attempt to re-acquire.

4 Build on Desktop Linux

```
make
```

Or manually:

```
g++ main.c -o tracker -I/usr/include/opencv5 \  
$(pkg-config --cflags --libs sdl2 SDL2_mixer) \  
-lraylib -lopencv_core -lopencv_videoio -lopencv_video \  
-lopencv_imgproc -lopencv_calib -lopencv_geometry -lopencv_tracking \  
-lGL -lm -lpthread -ldl -lrt -lX11
```

5 Build on Android / Termux

From a fresh Termux session, the one-shot installer handles everything (installs dependencies, and builds OpenCV from source if the Qt6 package breaks):

```
git clone <repo-url> tracker && cd tracker
bash build_termux.sh
```

Or step-by-step:

```
# 1. Enable x11-repo
pkg install -y x11-repo
# 2. Install dependencies
pkg update -y
pkg install -y clang make pkg-config opencv raylib sdl2 sdl2-mixer
# 3. Build
make
```

5.1 Termux runtime requirements

Component	Package	Purpose
Camera	termux-api	Access device camera from CLI
Display	termux-x11 or VNC	Render the window
Audio	Built-in	SDL2_mixer works out of the box

Display via termux-x11:

```
pkg install termux-x11
termux-x11 :0 &
export DISPLAY=:0
./tracker
```

Display via VNC:

```
pkg install tigervnc
vncserver :1 -geometry 1280x720
export DISPLAY=:1
./tracker
```

5.2 Camera on Android

Android does not expose `/dev/video0`. OpenCV's `VideoCapture(0)` requires either a USB webcam via OTG, or a device with camera access bridged through Termux.

6 Dependencies

Library	Version	Notes
Raylib	5.x-6.x	Windowing, rendering, input
OpenCV	4.x-5.x	core, video, videoio, imgproc, calib, geometry, tracking
SDL2 + SDL2_mixer	2.x	Audio playback and volume control

7 Troubleshooting

7.1 Qt6 / opencv package fails on Termux

Known issue: Termux Qt6 packages occasionally have dependency conflicts. The `build_termux.sh` script auto-detects this and builds OpenCV from source without Qt (15–25 min). The tracker does not use OpenCV GUI, so Qt is not needed.

Try first:

```
pkg upgrade -y
apt --fix-broken install -y
pkg install -y opencv
```

7.2 Build fails: "cannot find -lraylib"

Ensure x11-repo is enabled and raylib installed:

```
pkg install -y x11-repo raylib
```

7.3 Build fails: "cannot find -lGL"

Install OpenGL: `pkg install -y mesa`

8 Controls

Input	Action
Mouse drag / touch	Move nearest corner handle
S	Save calibration to <code>calib.txt</code>
R	Reset corners to full frame

9 Project Files

- `main.c` – source
- `Makefile` – build system (desktop + Termux)
- `build_termux.sh` – zero-dependency Termux setup script
- `remote_update.sh` – pull + rebuild on Android over SSH
- `calib.txt` – saved perspective calibration (gitignored)
- `samples/` – audio loops